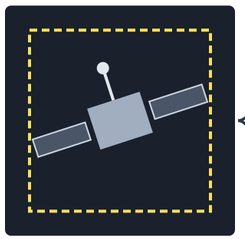
Two-branch hierarchical Tesseract predictor with PoE late fusion (Phase 3, EXP-036) and an image-conditioned SO(3) flow posterior (Phase 4, EXP-042)

frozen (no gradients) · trained, branch A (DINOv3) · trained, branch B (MoGe-2) · training-free fusion · Phase-4 flow (trained; trees frozen)

Phase 3 — discrete hierarchical prediction



SPEED+ image
square object crop (margin 0.15)
gray → 3 channels

DINOv3 ViT-L/16
self-supervised appearance features
❄ frozen

last layer, 16×16 → 8×8 pool

tokens 8×8
64 × 1024

MoGe-2 ViT-L
monocular geometry
surface normals + mask
❄ frozen

mask-weighted pooling

normals 16×16
256 × 4

Tesseract tree Transformer A

Encoder
LN + Linear → d = 256
+ 2-D positional enc.
1 self-attn layer
memory M_A

Autoregressive path decoder
ROOT · c_0 · c_1 · … · c_4
causal self-attn + cross-attn to M_A, 2 layers
cell token = level emb + Fourier(cell centre)
heads: root → 4 charts, child → 8

Tesseract tree Transformer B

same architecture, separate weights

Encoder
256 tokens → d = 256
memory M_B

Path decoder
p_B(chart), p_B(child | prefix)
per level, 6 levels (root + 5)

M_B (frozen)

log p_A^(l)

log p_B^(l)

Tesseract hierarchy of SO(3)



4 charts · 8 children / level

depth 5:
$4 \cdot 8^5 = 131{,}072$
leaf cells, ≈2.5°

q ≡ −q: positive facet of max $|q_i|$; cube → sphere by radial projection

Per-level PoE fusion

```
log p^(l) = log_softmax(
log p_A^(l) + log p_B^(l) )
```

equal weights, renormalised at every level · no trained parameters

greedy descent:
argmax chart, then argmax child ×5

Leaf-centre rotation q̂_tree

EXP-036 (3 seeds), mean error

synthetic val	**5.54° ± 0.06**
lightbox (real)	**26.9° (med 9.0°)**
sunlamp (real)	**36.6° (med 13.0°)**

≈0.4 ms / image (batched)

train: cross-entropy per level with geodesic soft targets (synthetic train only)
input features standardised with train statistics

Phase 4 — continuous posterior by Riemannian flow matching on SO(3) (EXP-042)

M_A (frozen)

R_τ, τ

Flow head A

query: Fourier(R_τ) ⊕ Fourier(τ)
MLP → 256, 2 cross-attn layers
→ ω_A(R_τ, τ | image)

Flow head B

same design, separate weights
cross-attends to M_B
→ ω_B(R_τ, τ | image)

Fused velocity

$\omega = \frac{1}{2}(\omega_A + \omega_B)$
heuristic, not an exact PoE

ω, every step

Euler on the group

$R \leftarrow R \cdot \exp(\Delta\tau \cdot \omega)$
τ: 0 → 1, Δτ = 0.1 (NFE = 10)
stays exactly on SO(3)
ω = body-frame velocity ∈ ℝ³

τ = 1

τ = 0

Base samples



R_0 ~ uniform (Haar) on SO(3)
M = 32 per image

Posterior samples



32 endpoints R_1 per image;
may hold a 180° flip mode (red)

q̂_flow + uncertainty

KDE mode (5° kernel); 90% region from sample spread

EXP-042 (3 seeds), mean error

synthetic val	**5.22° ± 0.15**
lightbox, KDE mode	**29.8°**
lightbox, best of 32 (oracle)	**9.7°**
coverage of 90% region	**0.98**

≈17 ms / image · only the flow heads are trained; backbones and trees stay frozen

Flow-matching training (synthetic train only)

$a = \mathrm{Log}(R_0^{-1} R_1^*)$, $R_\tau = R_0 \cdot \exp(\tau a)$, $\tau \sim U(0,1)$, 16 pairs / image

$L_{FM} = \|\omega_\theta(R_\tau, \tau \mid \text{image}) - a\|^2$ (geodesic path, constant target)

R_1^* = ground-truth rotation; pairs within 0.06° of the 180° cut locus are masked

Rotation only, ground-truth crops. Selection on synthetic validation; lightbox / sunlamp (real) are test-only. The real-domain gain comes from PoE (lightbox: DINOv3 alone 37.1°, MoGe-2 alone 43.7°, PoE 26.9°).
Remaining real-domain error is dominated by ~180° body-axis flips caused by visual near-symmetry (EXP-104); the flow's best-of-32 shows the correct mode is usually among its samples.